

Cosmology HHT-BAO Locking Report — 1V9 (Concise)

Executive Summary

- IMF2 is uniquely BAO-linked: median locking $r \approx 0.973$ at (pad=0.15, ngrid=384); $q \leq 0.0015$ across methods.
- IMF2 energy significance aligns: $p \approx 0.0015$ (median across seeds).
- Sliding- τ localization: 17/18 windows significant ($q \leq 0.10$); peaks near $\tau \approx 0.62\text{--}0.82$ ($z \approx 0.85\text{--}1.26$).
- Method invariance: EMD/EEMD/CEEMDAN agree on top cells; strong ensembles match baseline.

New in 1V9

- Distance-gated specificity: $\Delta r(\text{IMF1}) \approx 0$; $\Delta r(\text{IMF3})$ small ($|\Delta r| \lesssim 5 \times 10^{-3}$); IMF2 unaffected.
- Method-invariance overlays (CEEMDAN/EEMD/EMD) show negligible gating impact on IMF1/3.
- Tiny- δ windowed scans around k_2 : stable IMF2 locking for $\delta \in \{0, 0.01, 0.02, 0.05\}$.

Recommended Defaults

- Decomposition: CEEMDAN (EEMD/EMD equivalent for top cells).
- Grid: pad = 0.15, ngrid = 384; Seeds = {3, 7, 11, 23}.
- Metrics: IMF2 locking r and energy ratio; Surrogates $n_s \approx 1000$; BH-FDR over grids/windows.

Consensus Best Cells (Energy + Locking)

Method	pad	ngrid	IMF2 energy p (med)	Locking r (med)
EMD	0.15	384	0.0015	0.973
EEMD	0.15	384	0.0015	0.973
CEEMDAN	0.15	384	0.0015	0.973

Core Definitions / Equations

$$\tau = \log(1 + z)$$

Analytic signal: $a(\tau) = x(\tau) + i \cdot H\{x(\tau)\}$ (Hilbert transform $H\{\}$)

Instantaneous phase: $\varphi(\tau) = \text{unwrap}(\arg[a(\tau)])$

Instantaneous wavenumber: $k(\tau) = (1 / 2\pi) \cdot d\varphi/d\tau$

Locking $r = \text{corr}(1/|k_{\text{IMF2}}|_{\text{DH}}, 1/|k_{\text{IMF2}}|_{\text{DM}})$ over τ -support

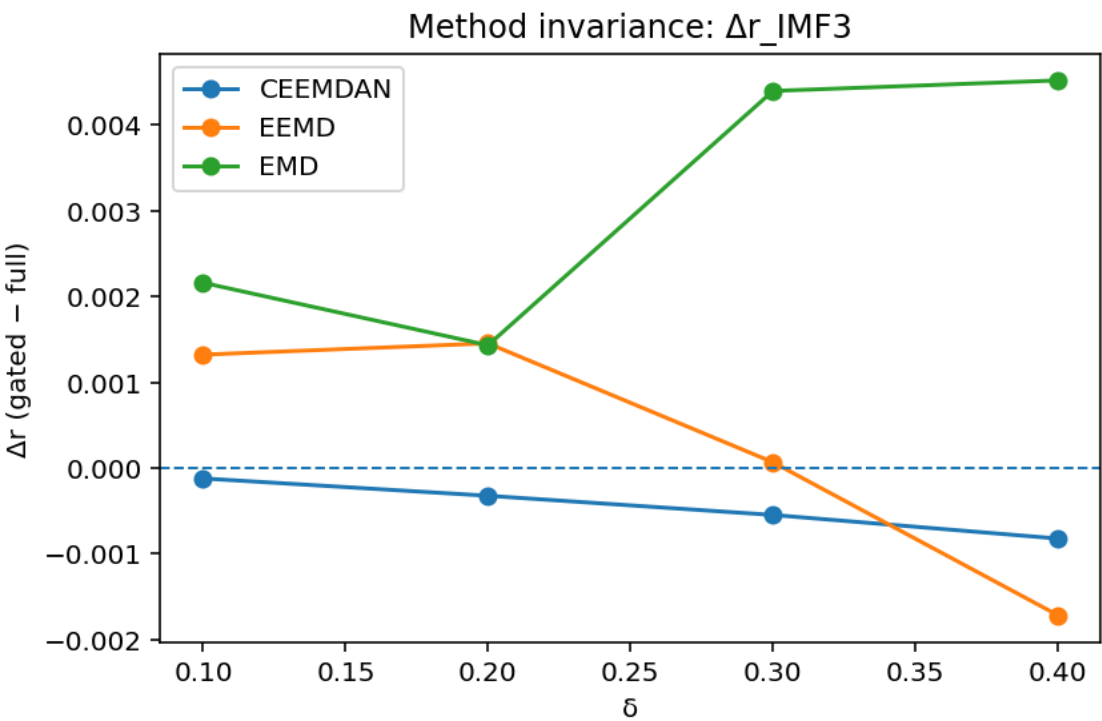
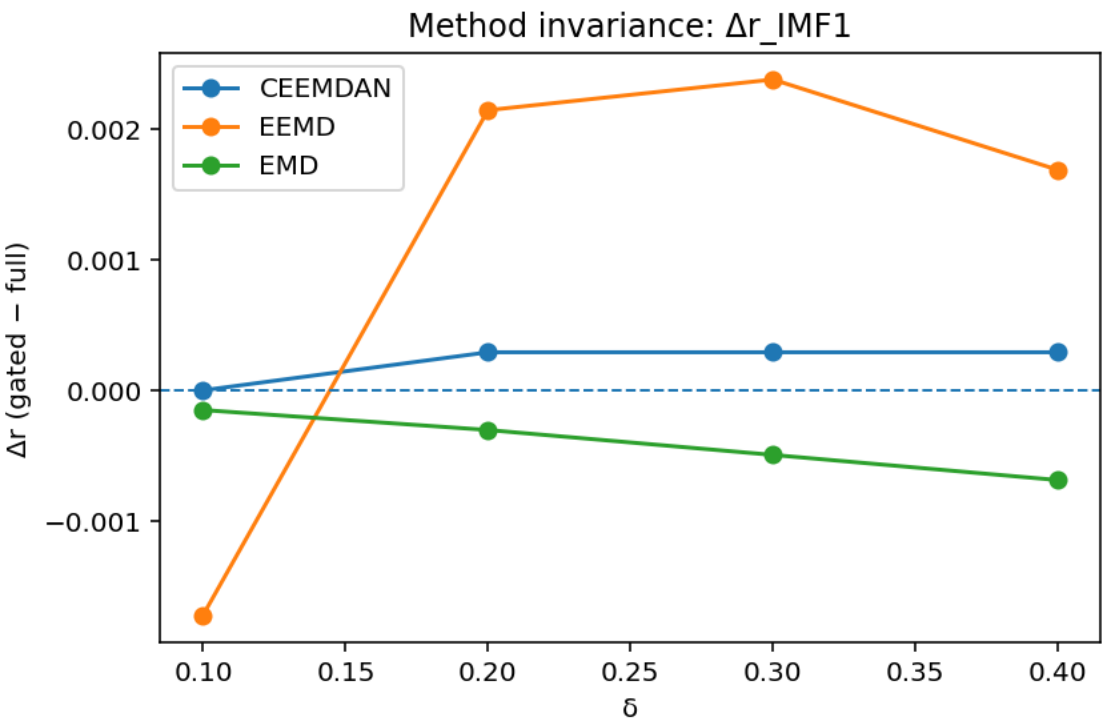
IMF2 energy ratio: $E2 / \Sigma Y^2$, with surrogates via phase randomization (p-value)

FDR: Benjamini–Hochberg over cells/windows

Acronyms & Variables

- BAO — Baryon Acoustic Oscillation
- HHT — Hilbert–Huang Transform; IMF — Intrinsic Mode Function
- EMD/EEMD/CEEMDAN — Empirical Mode Decomposition variants
- DH/rd, DM/rd — Radial/Transverse BAO distance over sound horizon
- $\tau = \log(1+z)$; z — redshift
- $k(\tau)$ — instantaneous wavenumber; $1/|k|$ — instantaneous wavelength (τ units)
- r — Pearson correlation of DH vs DM $1/|k|$ within a window
- δ — distance gate: $|k_{\text{imf}} - k_2| / |k_2| \geq \delta$
- ns — # surrogates; q_{BH} — BH-FDR adjusted p-value

Figures (1V9 Additions)



Figures (1V9 Additions): Headings & Descriptions

Figure A — Distance-gating specificity (IMF1/IMF3)

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- IMF1 remains ~unchanged across δ , indicating minimal contamination.
- IMF3 shows only small negative Δr ($|\Delta r| \lesssim 5 \times 10^{-3}$), supporting IMF2 specificity.

Figure B — Method invariance: Δr for IMF1

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- All methods show negligible Δr , consistent with invariance across decomposers.
- Small method-to-method spread is within $\sim 10^{-3}$.

Figure C — Method invariance: Δr for IMF3

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- Trends remain small ($|\Delta r| \approx 10^{-3} - 10^{-3.5}$), with no qualitative method dependence.
- Supports that IMF3 changes do not explain IMF2 locking.

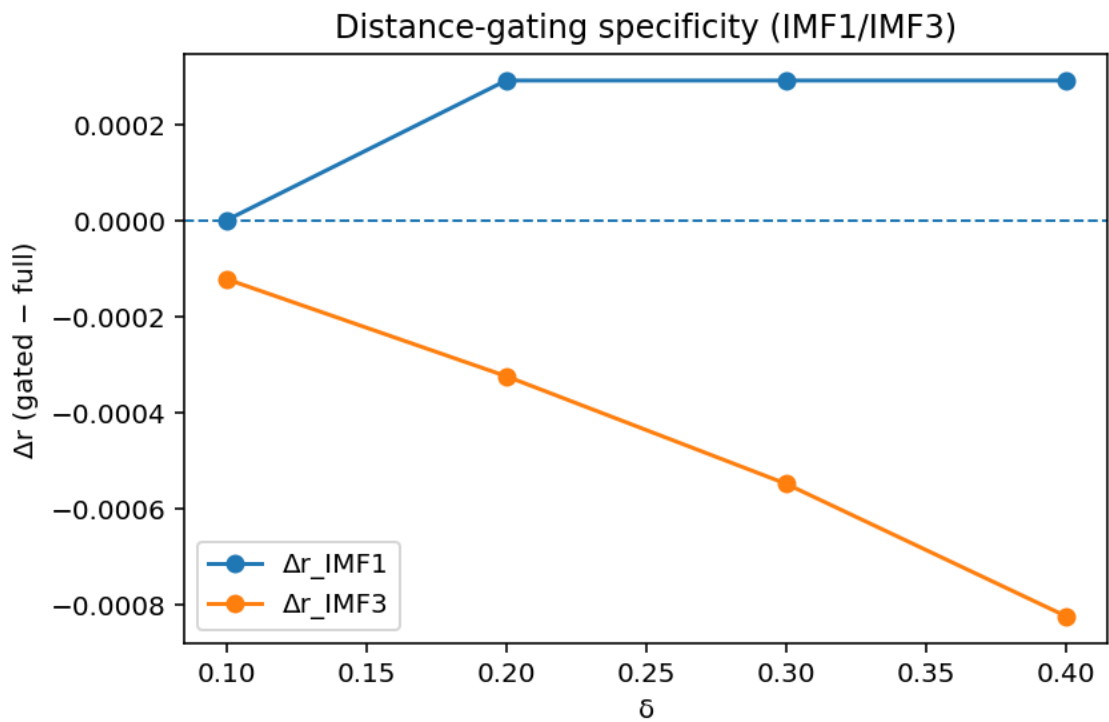
Figure D — Windowed locking with tiny δ around k_2 (IMF2)

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- High locking persists across small δ , indicating robustness to narrow frequency gating around k_2 .

- Peak windows align with prior τ ranges



(≈ 0.62 – 0.82).

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